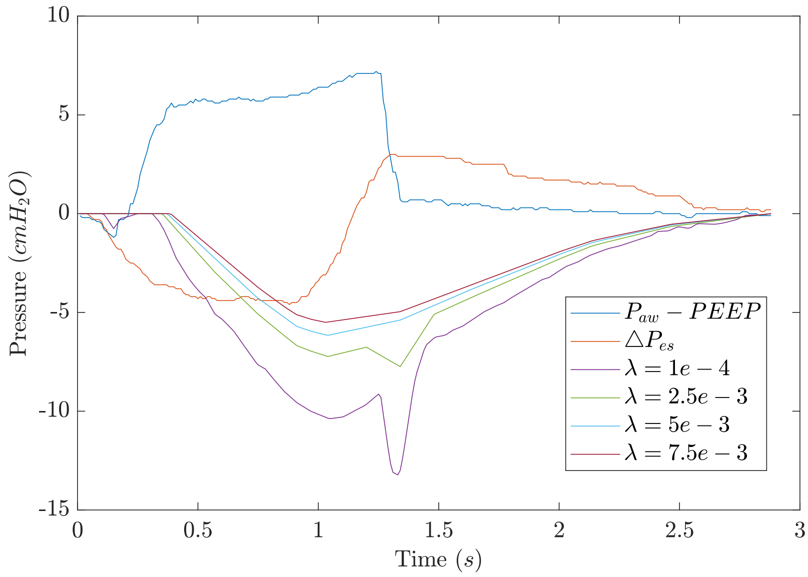
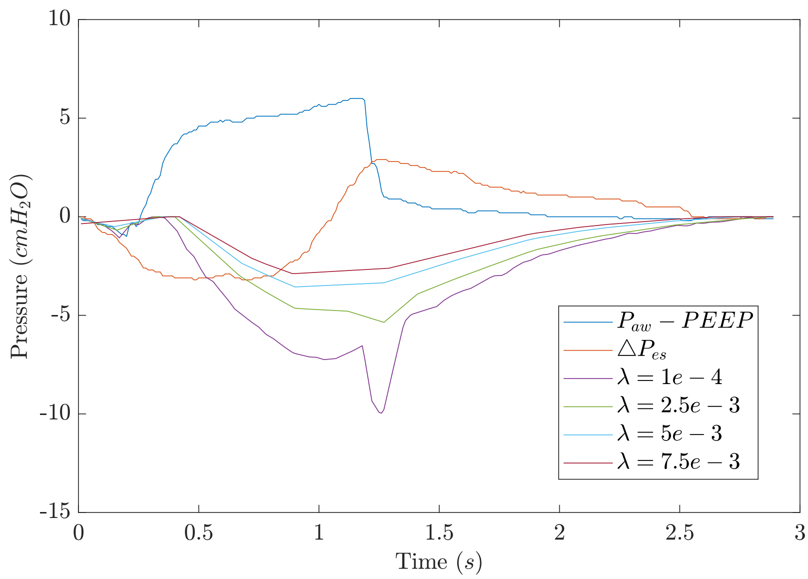
Appendix A: Other Methods

Sparse Optimisation

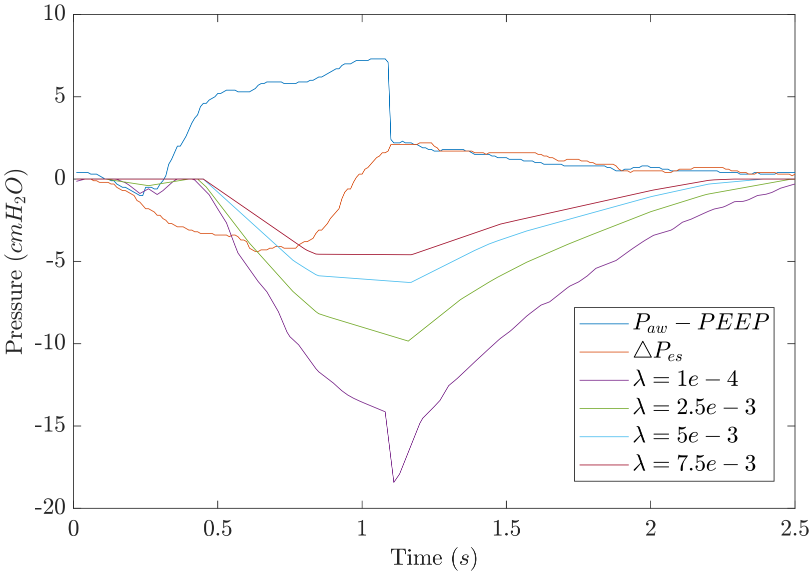
Patient 1a



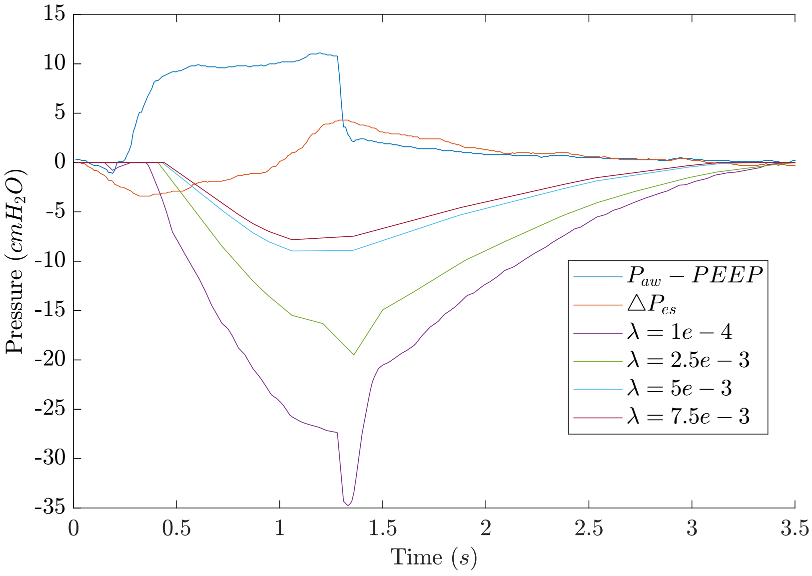
Patient 1b



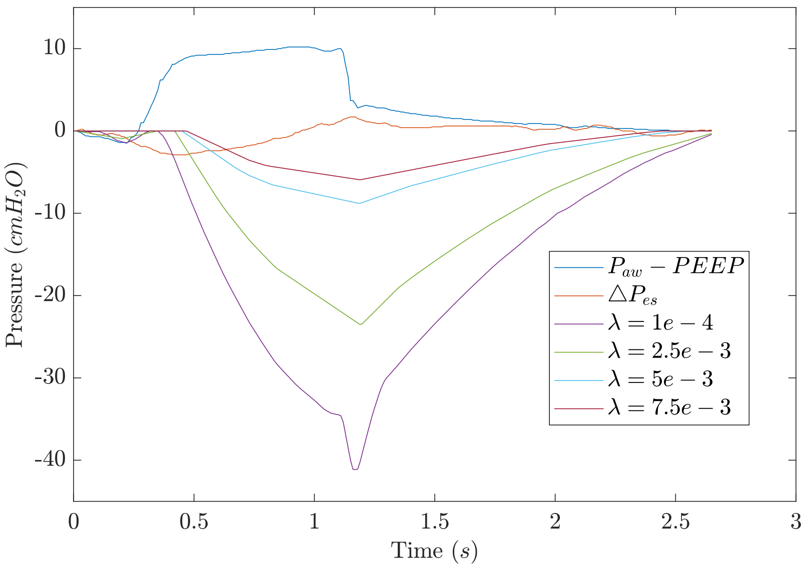
Patient 2a



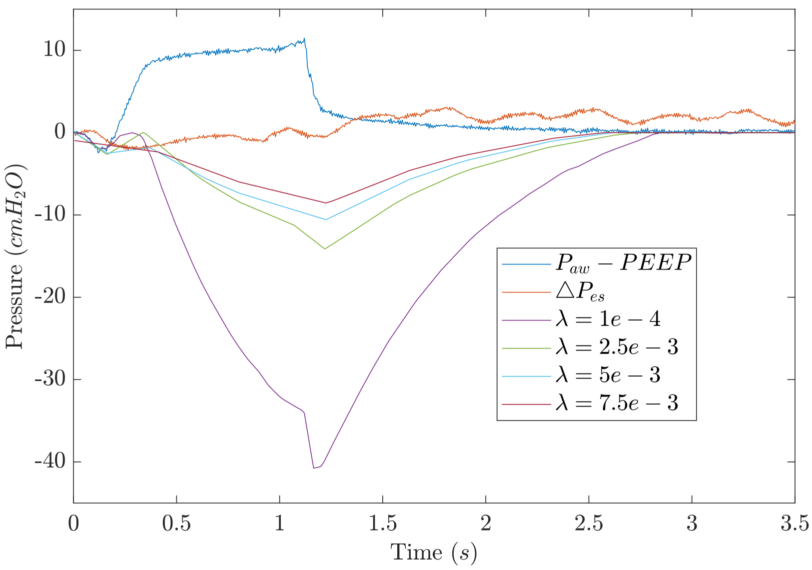
Patient 2b



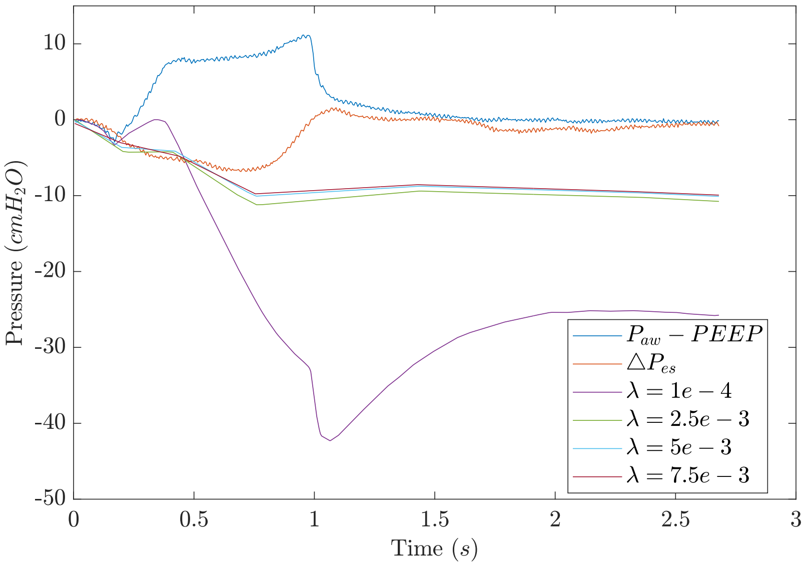
Patient 3



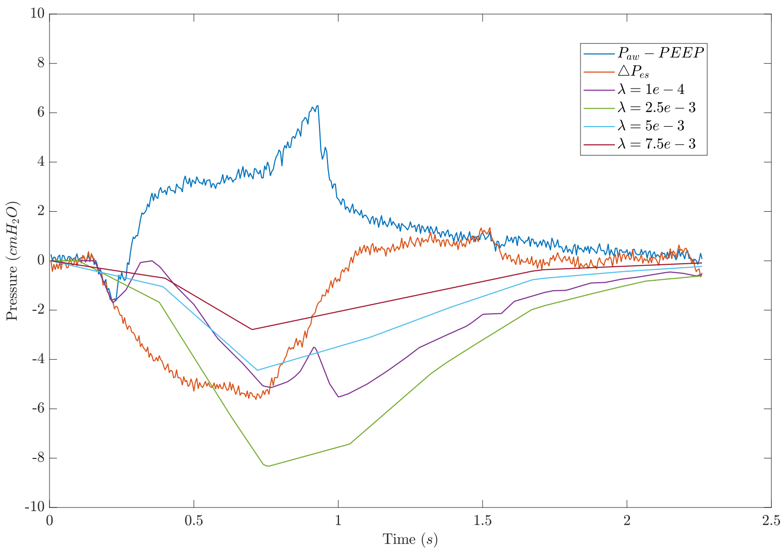
Patient 4



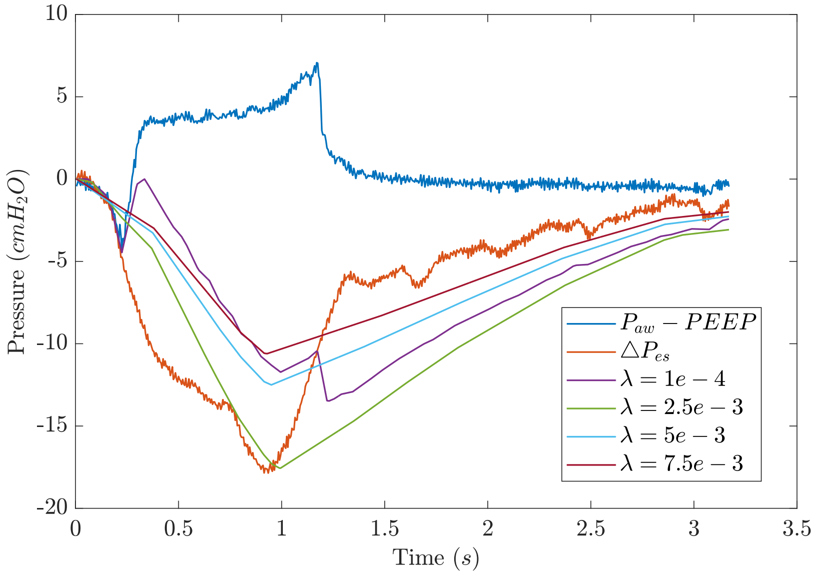
Patient 5



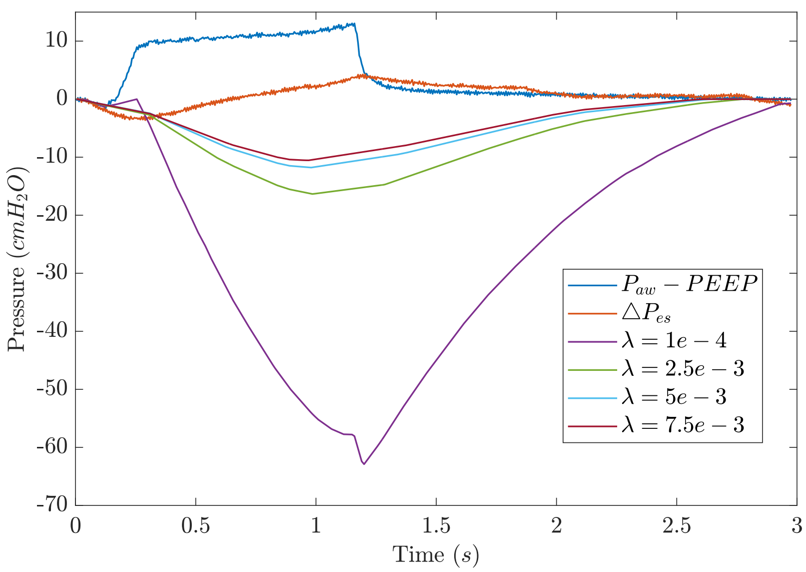
Patient 6



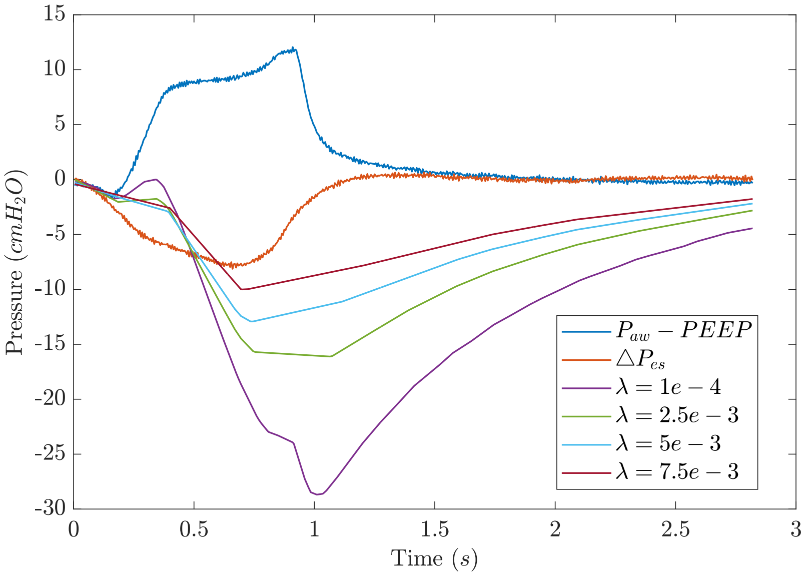
Patient 7



Patient 8



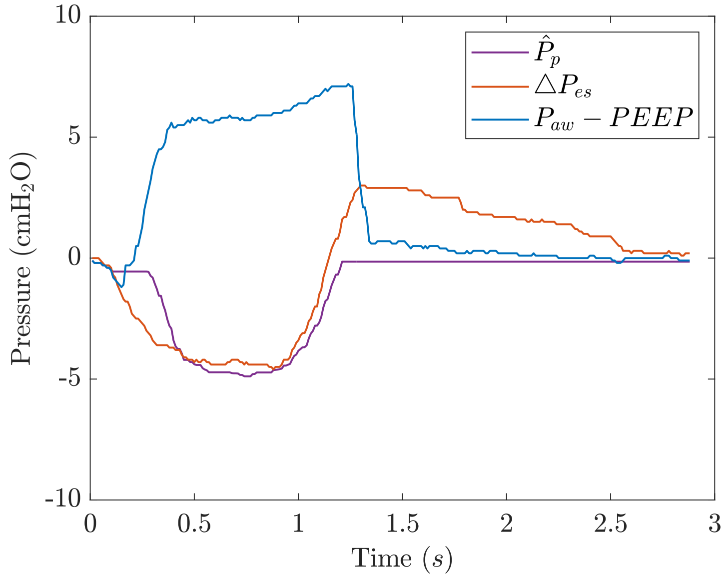
Patient 9



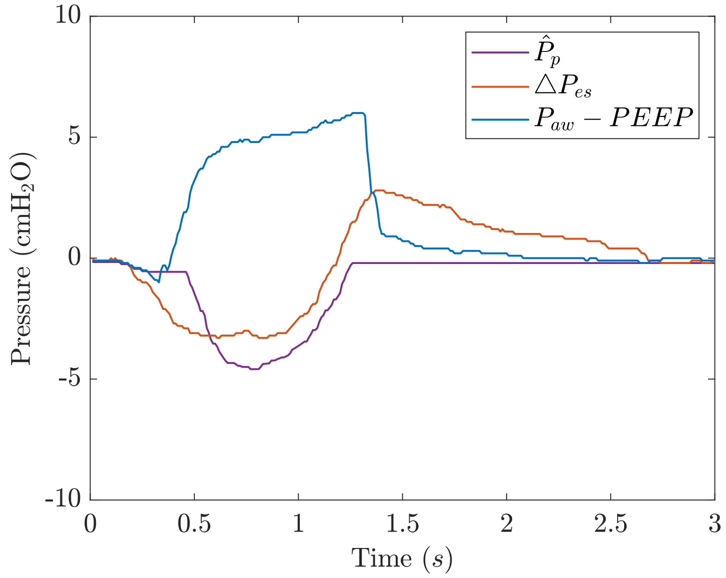
15
10
5
0
-5
-10
-15
-20
-25
-30
Pressure (cmH_2O)
0
0.5
1
1.5
2
2.5
3
Time (s)
$P_{aw} - PEEP$
$\triangle P_{es}$
$\lambda = 1e-4$
$\lambda = 2.5e-3$
$\lambda = 5e-3$
$\lambda = 7.5e-3$

Constrained Optimisation

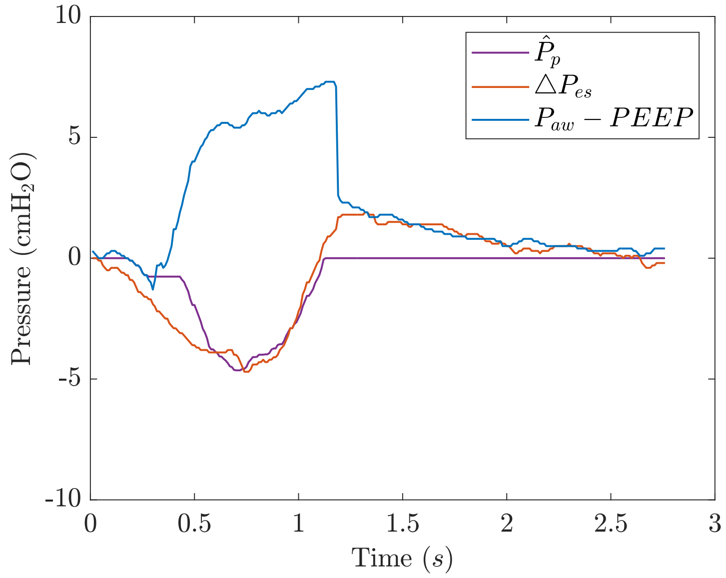
Patient 1a



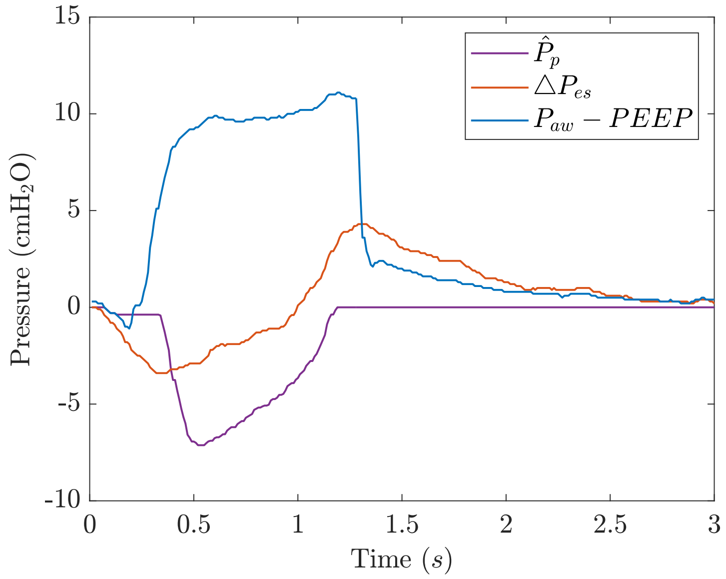
Patient 1b



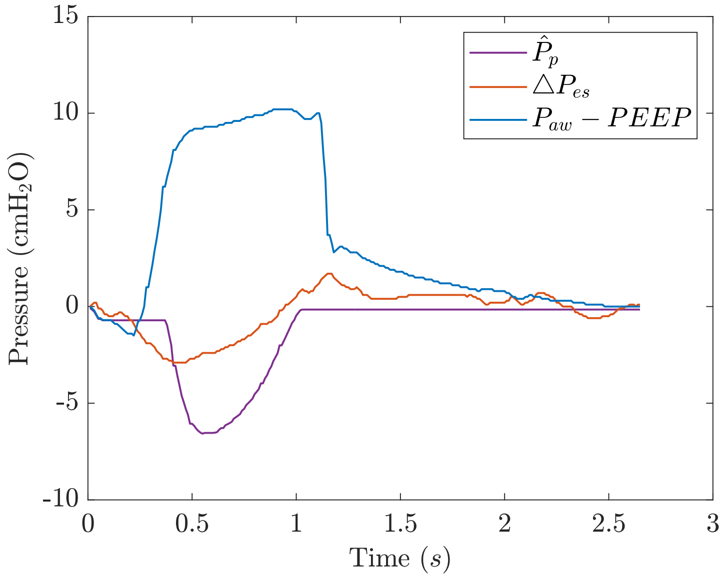
Patient 2a



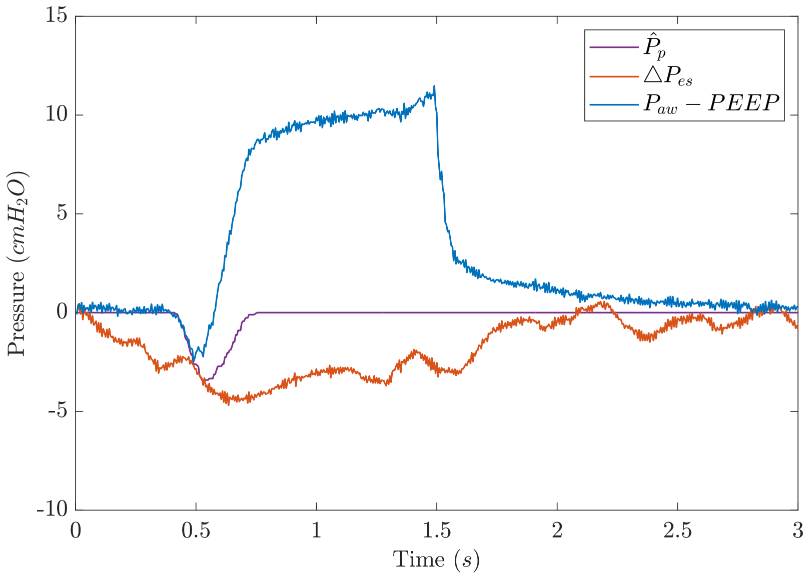
Patient 2b



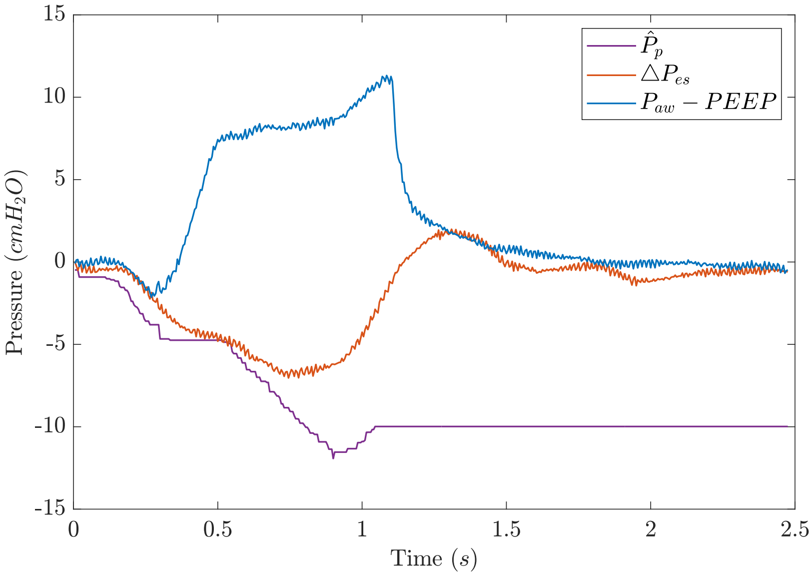
Patient 3



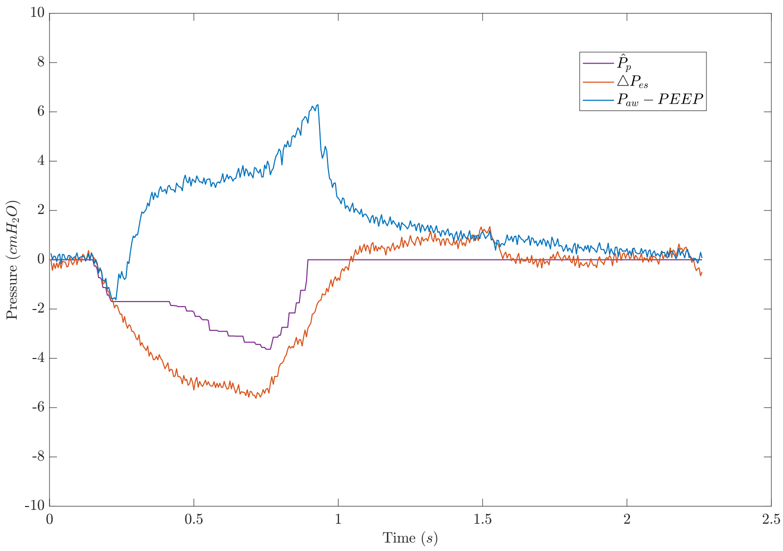
Patient 4



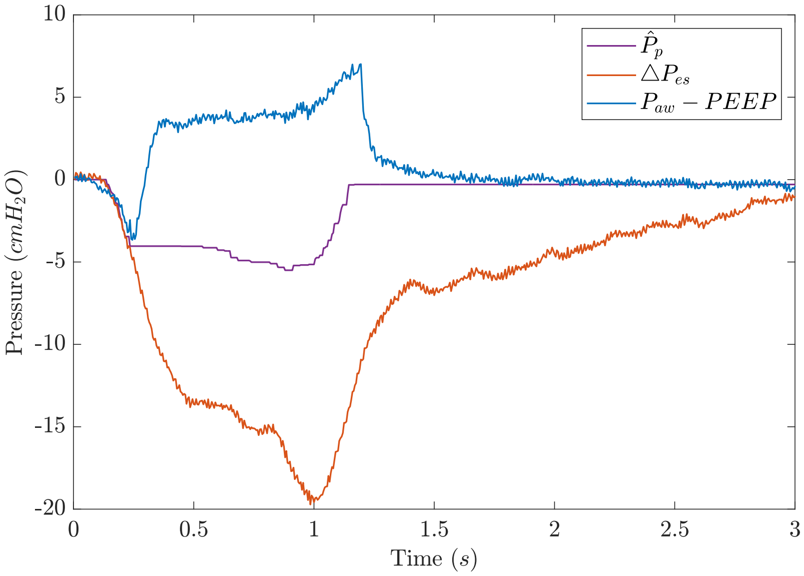
Patient 5



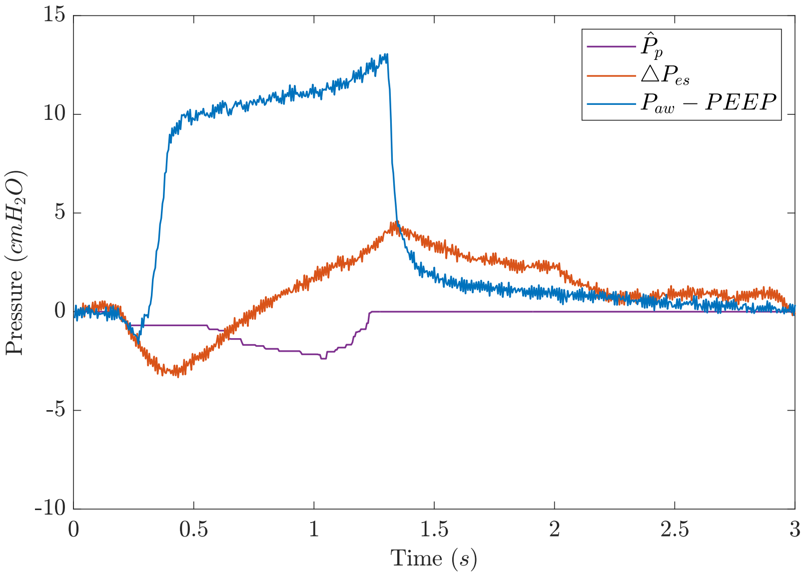
Patient 6



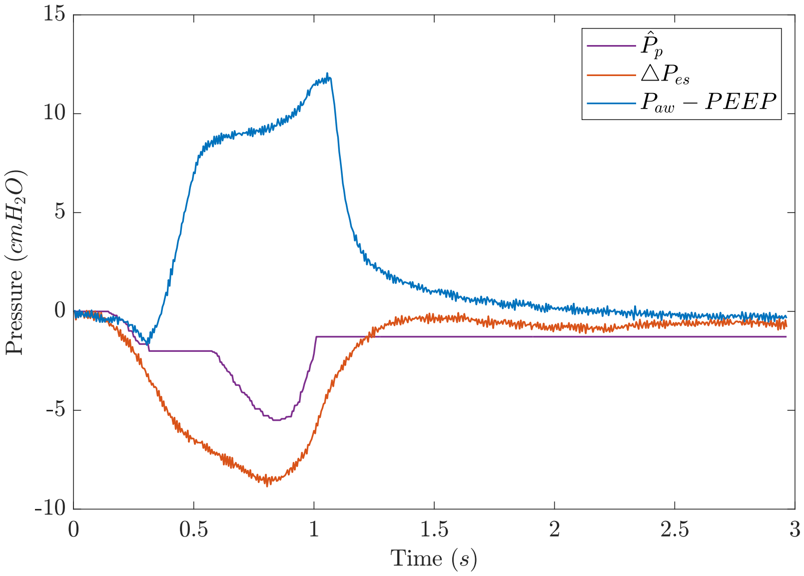
Patient 7



Patient 8



Patient 9



$\hat{P}_p$
$\triangle P_{es}$
$P_{aw} - PEEP$
Pressure (cmH_2O)
Time (s)
15
10
5
0
-5
-10
0
0.5
1
1.5
2
2.5
3

Appendix B: B-MIQP Model Framework

1. Model Description

A complete description of the B-MIQP model is provided. The B-MIQP model yields, via least squares estimation, constant estimates for a patient's respiratory mechanics over multiple breaths, and a unique waveform representing the pressure generated by the patient's inspiratory muscles ($\hat{P}_p$) for each breath. To achieve this, the constraints described in [20] have been reformulated to constrain the magnitudes of a set of basis-spline (b-spline) functions.

Airway pressure has previously been modeled using the single compartment lung model [21]. In this model, the pressure at the patient's airways (P_{aw}) is determined as the summation of: 1) airway flow ($\dot{V}$) and volume (V) signals scaled by respiratory system resistance (R) and elastance (E) values respectively, 2) a positive end-expiratory pressure ($PEEP$), used to maintain a minimum volume of air in the patient's lungs over the expiration phase, and 3) an additional pressure representing pressure generated by the patient's own respiratory efforts ($\hat{P}_p$).

A number of additions to this model have been considered in the B-MIQP model. First, separate respiratory mechanics for ventilator inspiratory ([R_{insp}, E_{insp}]) and expiratory ([R_{exp}, E_{exp}]) portions of the breath are considered, to account for known differences [27]. Second, $\hat{P}_p$ is described by the summation of a set of b-spline functions ($\phi_{j,2}$) scaled by corresponding spline coefficients (P_s). As such, the adapted single compartment lung model is given by:

$$\begin{aligned} P_{aw}(t) = PEEP + E_{insp}V(t)\big(1-\gamma(t)\big) + R_{insp}\dot{V}(t)\big(1-\gamma(t)\big) \\ +E_{exp}V(t)\gamma(t) + R_{exp}\dot{V}(t)\gamma(t) + \sum_{j=1}^{M} -P_{s,j}\Phi_{j,2}(t), \end{aligned} \tag{1}$$

where γ, represents a scalar binary indicator – equal to either 0 or 1 at each sampled point in time – ensuring inspiratory and expiratory mechanics are considered during ventilator inspiration and expiration phases respectively. The negative sign in front P_s reflects the convention that patient inspiratory breathing effort acts to decrease the pressure at the patient's airways.

The derivation of the b-spline curves has been outlined in Section 2.3.2 of the main text as per (6). Considering the B-MIQP model, T_{max} is given by the total length of time considered per model fitting, rounded up to the nearest integer of the spline knot width (k_w). Aligning with the B-spline model, $k_w = 0.05$ [s].

1.1. Model Constraints

Assumptions about the shape of $\hat{P}_p$ are incorporated into the model such that the only possible model solutions are the ones that yield an estimate of either muscle pressure (P_{mus}) or pleural pressure (P_{pl}). First, it is assumed that $\hat{P}_p$ will be first monotonically decreasing and subsequently increasing during inspiration, and absent during expiration. Second, we assume that the breathing effort shall finish prior to or soon after ventilator cycling.

To incorporate these assumptions into the model, three constraint regions have been defined which constrain successive P_s relative to one another.

$$\begin{aligned}
&\text{Region 1: } P_{s,i,j+1} + \epsilon \geq P_{s,i,j}, \\
&\text{Region 2: } P_{s,i,j+1} \leq P_{s,i,j} + \epsilon, \\
&\text{Region 3: } P_{s,i,j} = 0,
\end{aligned} \tag{2}$$

where ϵ is a small constant ($\epsilon = 1e - 3$) to drive a minimum variation between successive values in Region 1 and 2. Combined, the three regions enforce the assumption that a breathing effort will be first monotonically decreasing and then monotonically increasing during an inspiratory effort, and that expiration occurs passively. The boundaries between regions for each breath are determined using switching point binary decision variables ($b_{i,j,k}$). The value of $b_{i,j,k}$ (either 0 or 1), is determined by the model within a mixed integer framework.

For each b-spline used to describe each breath, relevant $b_{i,j,k}$ are evaluated against a set of inequalities, to determine which regional constraint in (1) to enforce. Considering the ℓ-th b-spline of breath i, the following inequalities are evaluated:

$$\begin{aligned}
&\sum_{j=1}^{\ell} b_{i-1,j,3} - \sum_{j=1}^{\ell} b_{i,j,1} \geq 0.5 \implies \{\text{Region 3}\}, \\
&\sum_{j=1}^{\ell} b_{i,j,1} - \sum_{j=1}^{\ell} b_{i,j,2} \geq 0.5 \implies \{\text{Region 1}\}, \\
&\sum_{j=1}^{\ell} b_{i,j,2} - \sum_{j=1}^{\ell} b_{i,j,3} \geq 0.5 \implies \{\text{Region 2}\}, \\
&\sum_{j=1}^{\ell} b_{i,j,3} - \sum_{j=1}^{\ell} b_{i+1,j,3} \geq 0.5 \implies \{\text{Region 3}\}, \forall 1 < i < N_b
\end{aligned} \tag{3}$$

for $b_{i,j,k} i = 2, \ldots, N_b - 1$, $j = 1, \ldots, M_i$, and $k = 1, \ldots, N_s$. Three switching points per breath are considered (i.e., $N_s = 3$ for all i). For breath $i = 1$, inequality

1 is instead evaluated as $\sum_{j=1}^{\ell} b_{1,j,1} \leq 0.5 \implies \{\text{Region 3}\}$. Similarly, for breath $i = N_b$, inequality 4 is instead evaluated as $\sum_{j=1}^{\ell} b_{N_b,j,3} \geq 0.5 \implies \{\text{Region 3}\}$.

The time location of the k-th switching for point for breath i can be recovered as:

$$T_{s,k,i} = \sum_{j=1}^{M_i} t_{peak,i,j} \cdot b_{i,j,k}, \tag{4}$$

where $t_{peak,i,j}$ represents a vector containing the points in time where the peak of each b-spline function used to describe breath i occurs. Of note, $\phi_{i,j,2}$ and $t_{peak,i,j}$ are arranged such that switching points align with peaks of b-spline functions.

The locations of T_s are constrained relative to ventilator triggering and cycling. For each breath i, the location of the inspiratory switching point ($T_{s,insp}$, $k = 1$) is constrained to occur to or at the point of ventilator triggering (τ_{soi}) within $k_{soi} = 0.6$ [s], such that:

$$\tau_{soi,i} - k_{soi} \leq T_{s,insp,i} \leq \tau_{soi,i}. \tag{5}$$

Similarly, the expiratory switching point for a given breath ($T_{s,exp}$, $k = 3$) must occur within $k_{soe} = 0.25$ [s] after the point of ventilator cycling (τ_{soe}) for breath i.

$$T_{s,exp,i} \leq k_{soe} + \tau_{soe,i}. \tag{6}$$

As outlined in [23], k_{soi} were chosen from visual inspection of the onset of patient effort and ventilator triggering, considering measured P_{es} signals. Further, the value of k_{soe} aligns with that of the Mixed-Integer Quadratic Programming model described in [20].

Finally, switching points are constrained such that they must occur sequentially. As such, for a given b-spline ℓ, the following two inequalities hold true.

$$\sum_{j=1}^{\ell} b_{i,\ell,2} - \sum_{j=1}^{\ell} b_{i,\ell,1} \leq 0 \; \forall i \forall \ell \in \{1, \ldots, M_i\}, \tag{7}$$

$$\sum_{j=1}^{\ell} b_{i,\ell,3} - \sum_{j=1}^{\ell} b_{i,\ell,2} \leq 0 \; \forall i \forall \ell \in \{1, \ldots, M_i\}. \tag{8}$$

The model and optimization problem were formulated using the YALMIP framework (Taipei, Taiwan) within the MATLAB environment (Mathworks, Natick, MA, USA). The optimization problem was solved using the Gurobi MIQP solver (Houston, TX, USA). The Gurobi solver tolerance settings are listed in Table 1. All settings not listed in the table were set to their default value.

Table 1: Gurobi/Yalmip Model Settings. All unspecified variables, set to default variable

Setting	Value
Verbose	1
Lift Project Cuts	1
Method	0

1.2. Model Parameter Sensitivity Analysis

1.2.1. Number of Breaths

Table 2 lists the model outcomes from B-MIQP model fittings considering Patient 1a where the number of breaths considered in a single model fitting (N_b) are increased from N_b = [1, 2, 5, 10, 20]. 20 breaths are considered in total for each N_b considered. As an example for $N_b = 5$, 4 model fittings are considered, ensuring analysis of all 20 breaths. First, as observed in Table 2, as N_b increases the standard deviation in Elastance ($\sigma(E_{insp})$) and Resistance ($\sigma(R_{insp})$) appears to decrease. The relationship between $\sigma(E_{insp})$ and N_b was statistically significant (p-value = 0.025), while the equivalent relationship considering $\sigma(R_{insp})$ was not (p-value = 0.098). Further, as N_b increases, model solve times also increase, where notably, a significant increase in solve times is observed comparing $N_b = 10$ and $N_b = 20$. The standard deviation between differences in measured and modeled peak efforts ($\sigma(\triangle P_{es,\,peak} - \hat{P}_{p,\,peak})$) did not appear to significantly decrease with increases in N_b. The standard deviation in peak measured effort ($\sigma(\triangle P_{es,\,peak})$) across 20 breaths was relatively low ($\sigma(\triangle P_{es,\,peak})$) = 0.503 cm$H_2O$). Overall, these results suggest that considering a greater number of breaths per model fitting tended to decrease variation in respiratory mechanics estimates between model fittings while increasing model solve times. The influence of N_b on the accuracy of the patient effort estimates and the ability of the model to track changes in effort between breaths is likely dependent on the extent to which the patient's efforts vary between breaths. In this study, $N_b = 10$ was considered to minimize variation in respiratory mechanic estimates that was assumed non-physiological, whilst maintaining model solve times that remain tractable for bedside implementation. $N_b > 20$ was not considered as such considerations have previously been shown to not improve the performance of the B-Spline model [22].

1.2.2. Knot Width

Similar sensitivity analysis was performed considering the spline knot width (k_w) (results shown in Table 3). Considering Patient 1a, and $N_b = 10$, multiple B-MIQP model fittings of the same set of breaths were considered where k_w was increased from k_w = [0.05, 0.1, 0.2, 0.5] seconds. The results highlighted that as k_w was increased, model solve times decreased at a compromise of model accuracy, evaluated as the mean difference between measured ($\triangle P_{es}$) and modeled ($\hat{P}_p$) peak values. The same

Table 2: Performance of B-MIQP considering different number of breaths (N_b) per model fitting.

Number of Breaths	Solver Time Range	$\sigma(E_{insp})$	$\sigma(R_{insp})$	$\sigma(\triangle P_{es,\,peak} - \hat{P}_{p,\,peak})$
1	0.421-1.21	1.10	0.674	0.825
2	0.657-1.96	0.683	0.274	0.68
5	2.74-7.87	0.557	0.273	0.741
10	43.6-210	0.338	0.084	0.761
20	1.36e4	0	0	0.742

Table 3: Performance of B-MIQP model considering different knot width (k_w) values.

k_w [s]	Trial	Solver Time [s]	Mean Difference Peaks [cmH_2O]
0.05	1	37.7	1.24
	2	37.8	1.24
	3	38.3	1.24
0.1	1	8.45	1.50
	2	8.41	1.50
	3	8.79	1.50
0.2	1	1.70	1.64
	2	1.68	1.64
	3	1.74	1.64
0.5	1	0.404	3.76
	2	0.377	3.76
	3	0.362	3.76

model fitting was performed three times per k_w value, showcasing that solver times remained consistent and the model appears to converge to the same solution for each fitting. $k_w = 0.05$ was selected in this study to optimize model performance whilst maintaining computational tractability.